

Facilitating zero-shot in Sim-To-Real Transfer of Avian Informed Drone

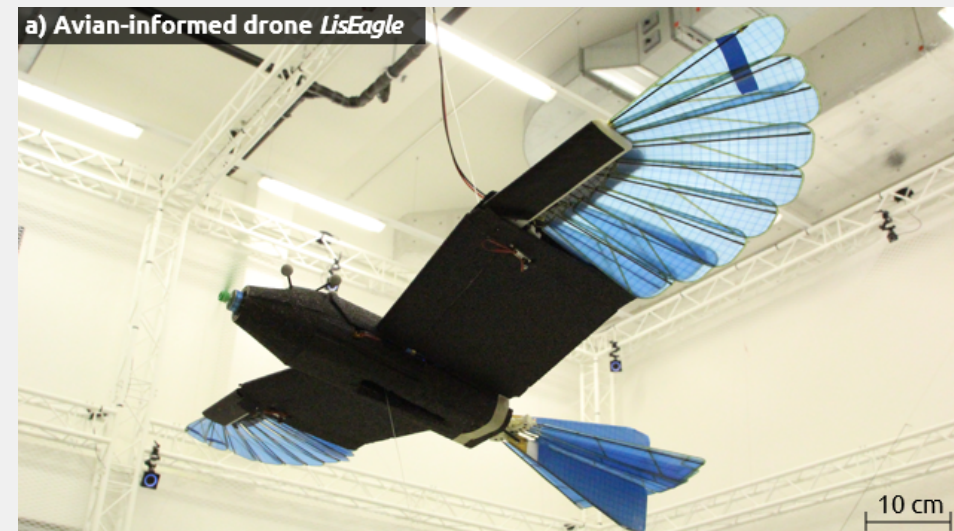
Victor Labbe, Erwann de Belloy, Alexandre Vray, supervised by Simon Jeger

EPFL

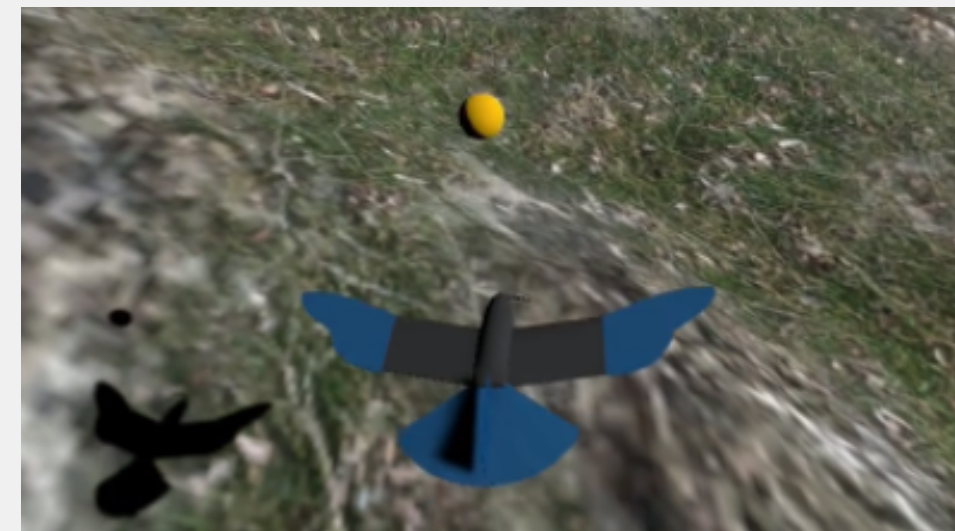
Avian Informed Drones

Introduction to LisEagle

The Laboratory of Intelligent Systems at EPFL is the birthplace of LisEagle, a drone that draws inspiration from avian flight. This bioinspired drone was initially steered by manual controls [2] [3] [4]. The next evolutionary step for LisEagle is to achieve autonomous flight, a goal that we are pursuing through the implementation of reinforcement learning techniques.



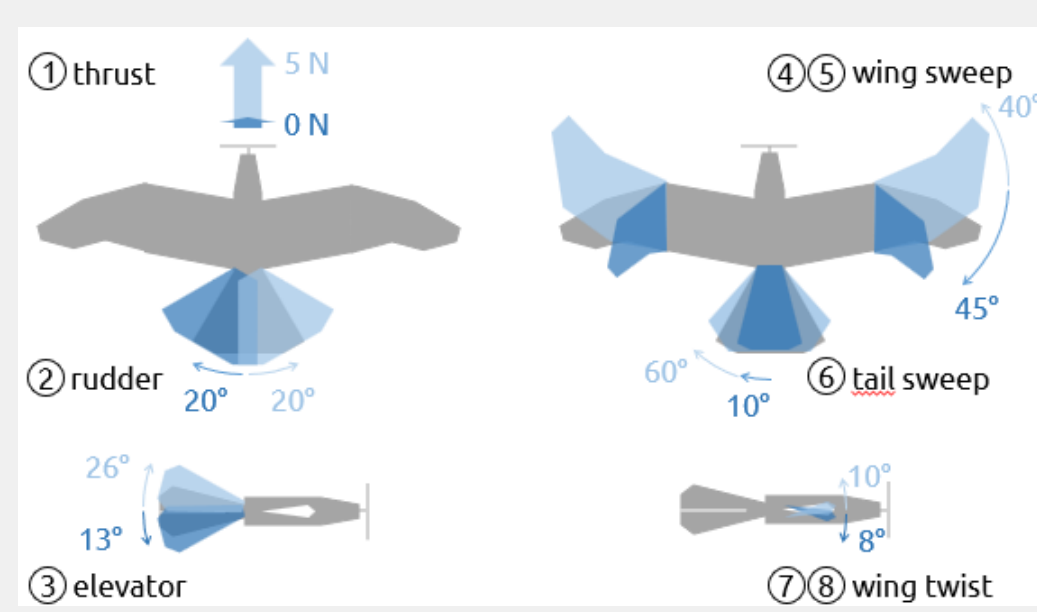
(a) LisEagle Drone flying



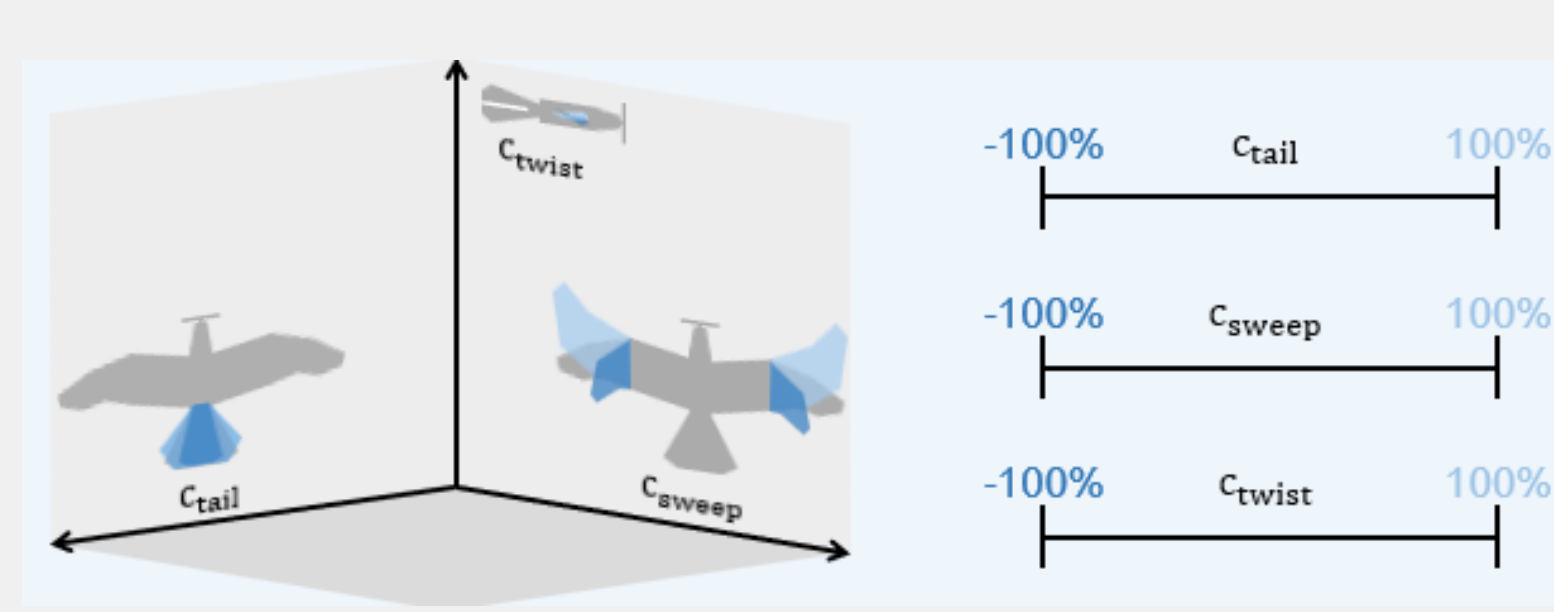
(b) Flight simulation in a Flightmare environment

Reinforcement Learning Implementations

Research Focus: Comparing Two Reinforcement Learning Implementations for the control of the LisEagle.



(a) Degrees of freedom



(b) Morphing configuration to improve energetic efficiency

State Space: 12 Degrees of Freedom (Position, Velocity, Attitude, Angular Velocities)

Action Spaces:

- End-to-end Control (8DoF):** Includes wings and tail sweep, wings twist, thrust, elevator and rudder on the tail (Fig.7a)
- Cascaded Control (7DoF) via an INDI Controller:** RL network as an abstraction layer outputs thrust, desired body rates, and wings/tail configurations, tracked by an Incremental Non-Linear Dynamic (INDI) controller. This approach aims at improving the robustness of the flight and facilitate simulation to real transfer (Fig.7b)

Objective: Assess the performance and robustness of both control strategies.

Algorithm: Proximal Policy Optimization (PPO) from Stable Baseline 3

Methodology:

- Metrics:** Establish metrics for quantitatively assessing model performance and robustness.
- Environment:** The LisEagle aims for a target point while facing variable front wind.
- Scenarios:** First a simplified uniform wind with stochastic variations around a nominal speed to validate our implementations. Then a more realistic wind model with von Karman turbulences [1].
- Reward:** Our reward is a linear combination of log-distances to target, yaw and orientation penalties, velocities and acceleration penalties, and an energy penalty. In the case of INDI RL trainings, a penalty for delays in the low level controller is added but removed for testing.
- Evaluations:** Compare both models in terms of reward and robustness metrics under varied environmental conditions. Robustness is hard to define, we choose to assess it through mean episode length, and neuron coverage and a sensitivity analysis. [6] [7][5]

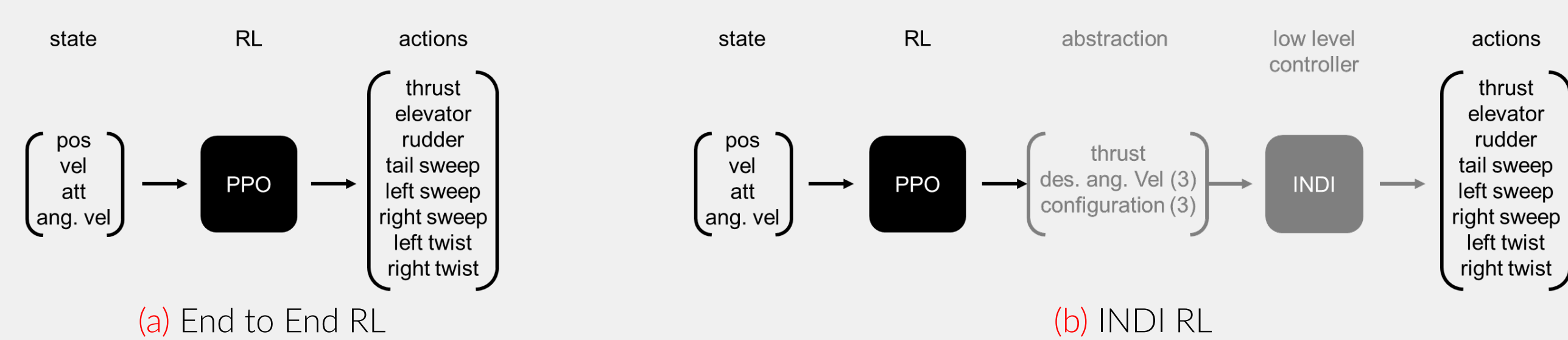
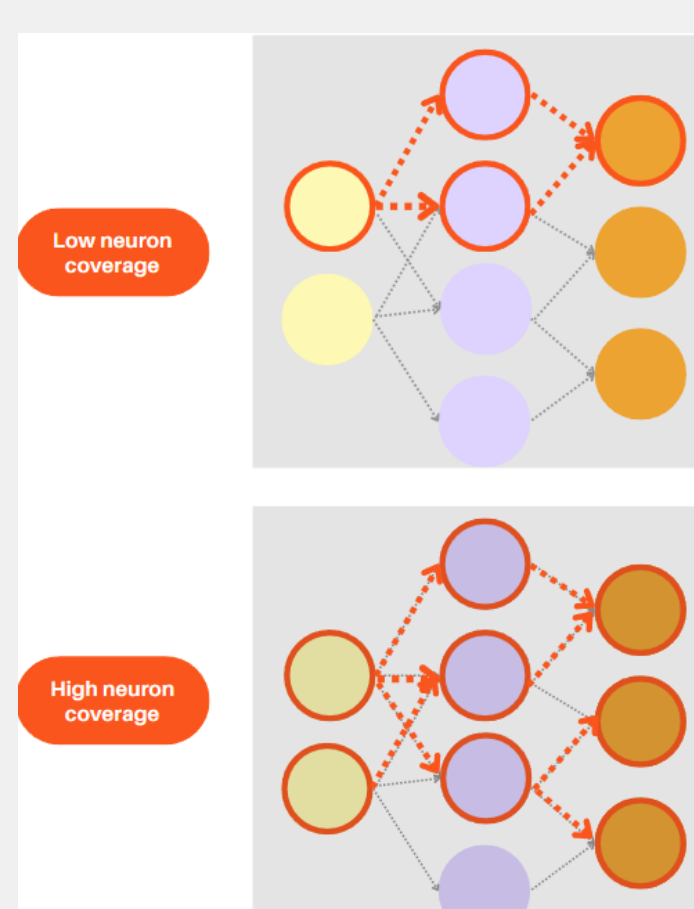


Figure 3. State to Action RL control with and without a low-level controller

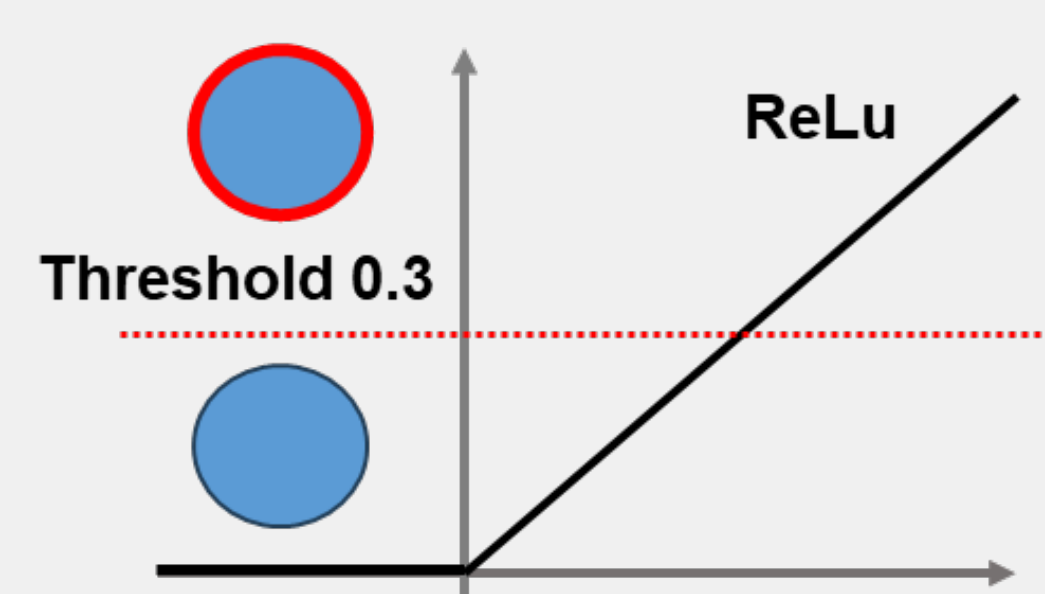
Toward Robustness with Neuron Coverage

Neuron coverage is a metric that can be used to assess the robustness of neural networks. It is a measure of the proportion of neurons in a network that are activated in the forward pass of the network by a given input. This concept have been initially introduced in [5] to generate augmented dataset that achieve better robustness for similar performances.

To implement a Neuron Coverage measure for the LisEagle, we introduced a callback mechanism within the neural network to capture each neuron's output. This allowed us to monitor neuron activation patterns under various perturbation scenarios. We fixed empirically a threshold of 0.3 for the neuron activation.



(a) Networks with different neuron coverage



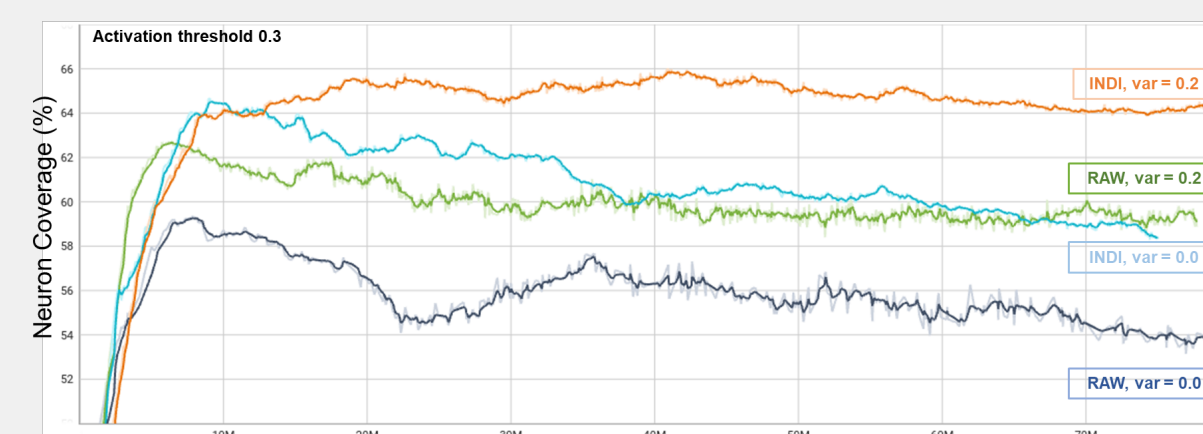
(b) Neuron Activation

Comparison of End-to-end RL and INDI RL controllers

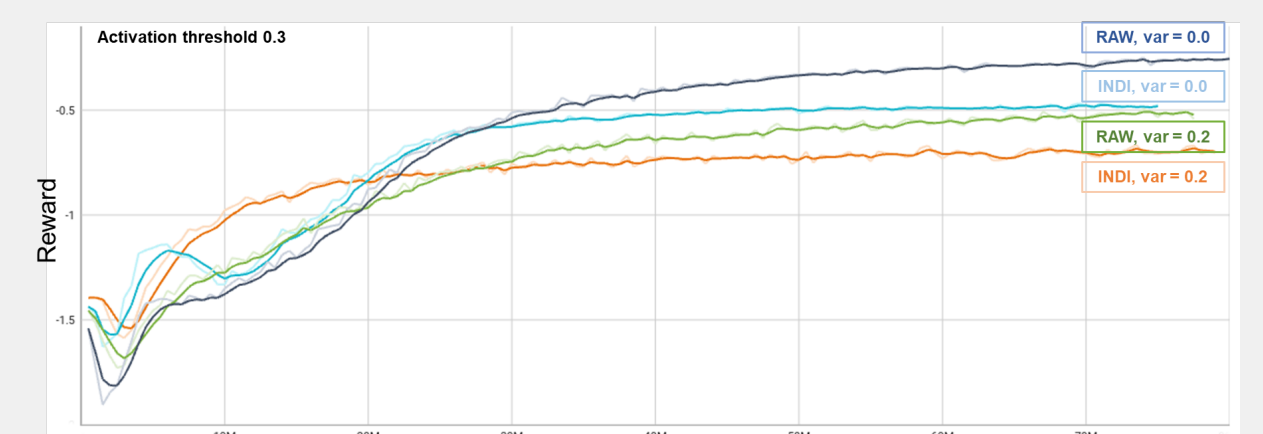
Evolution of Neuron Coverage and Reward during training

At every beginning of training, Neuron Coverage rises significantly. Then, as the policy converge and the reward stabilizes, neuron coverage seems to settle to a fixed value. However it is interesting to notice that for policies trained with low-to-no perturbations, the neuron coverage slowly decreases. We interpret this as an indication that the network might be over-fitting.

We notice an expected decrease of the final reward when we increase perturbation during training. Nonetheless, the difference between INDI and End-to-end final rewards is roughly equal to the low-level delay contribution in INDI reward. This would seem to indicate that the abstraction layer yield similar performances than the end to end implementation.



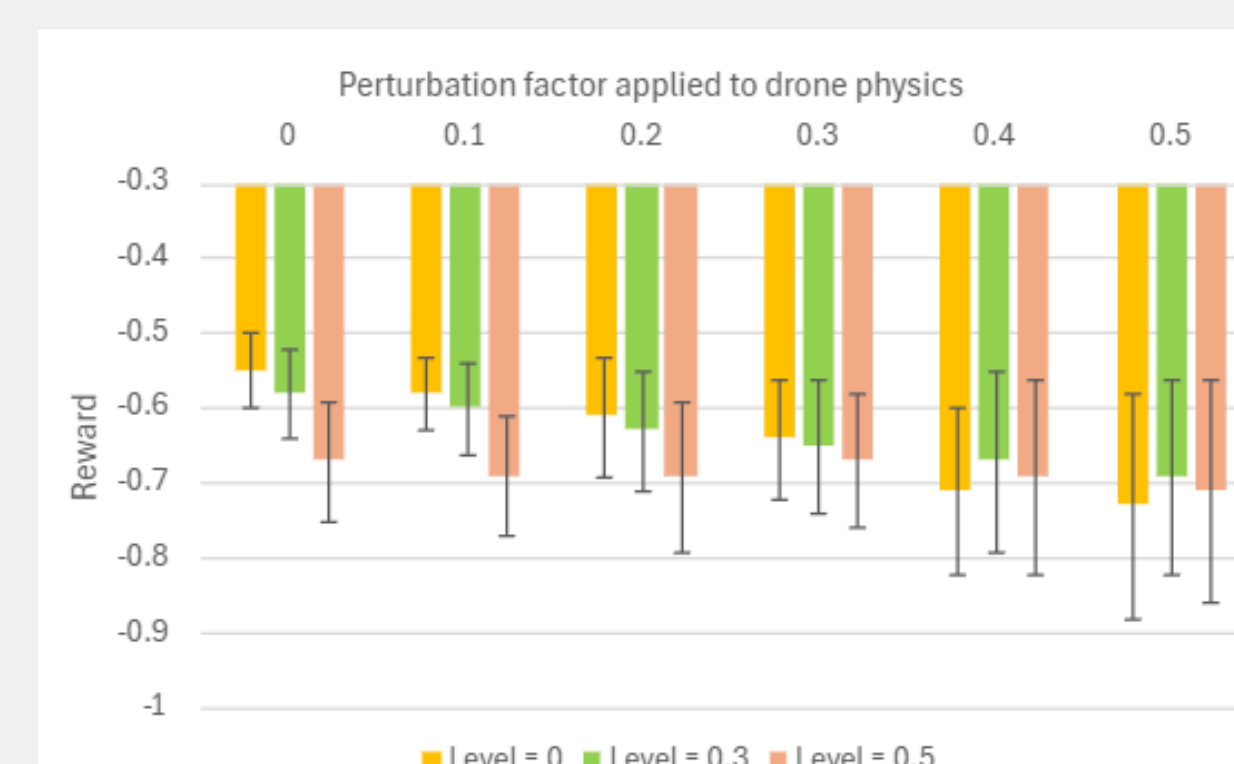
(a) Evolution of Neuron Coverage



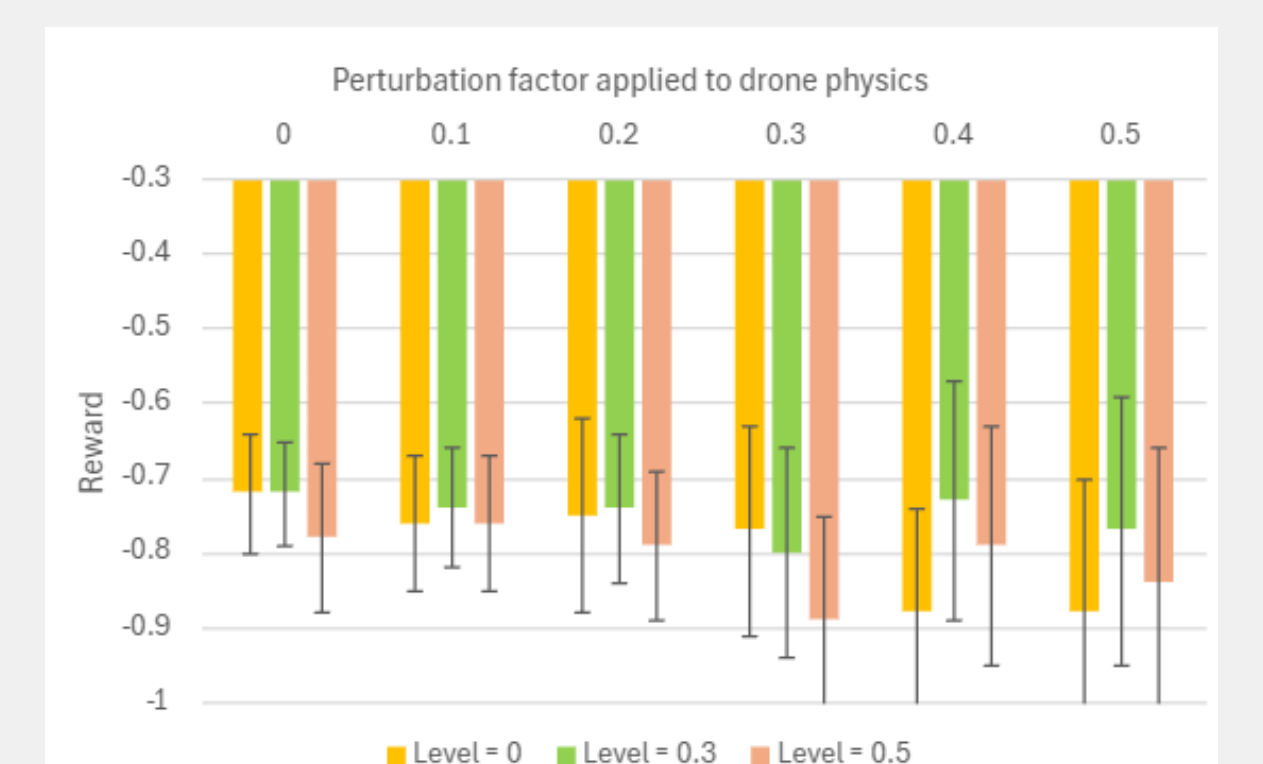
(b) Evolution of Reward

Comparison of Neuron Coverage and Reward during testing

We test 3 models trained with various level of noise for each approach. The noise is here a random perturbation proportionally applied to the drone's physics. This noise is gradually increased to 50% in the testing and the reward and NC are measured over 100 episodes.

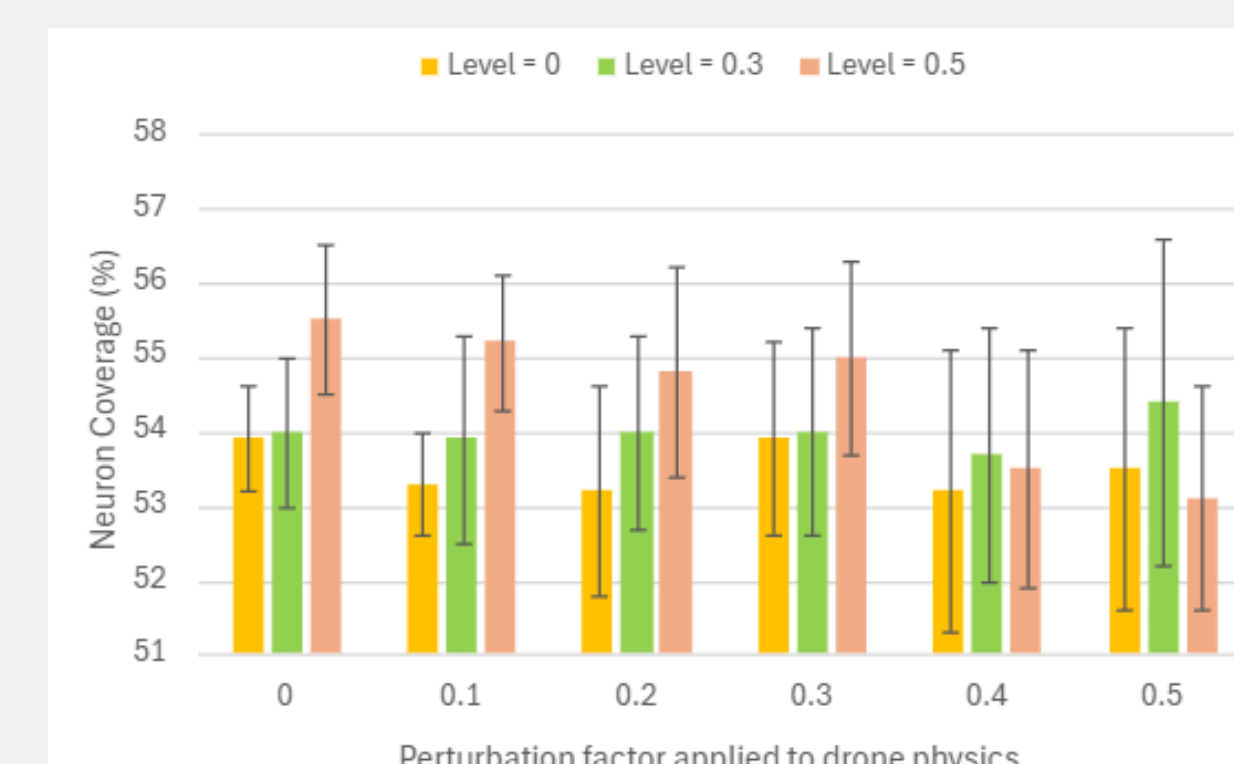


(a) End-to-end controller

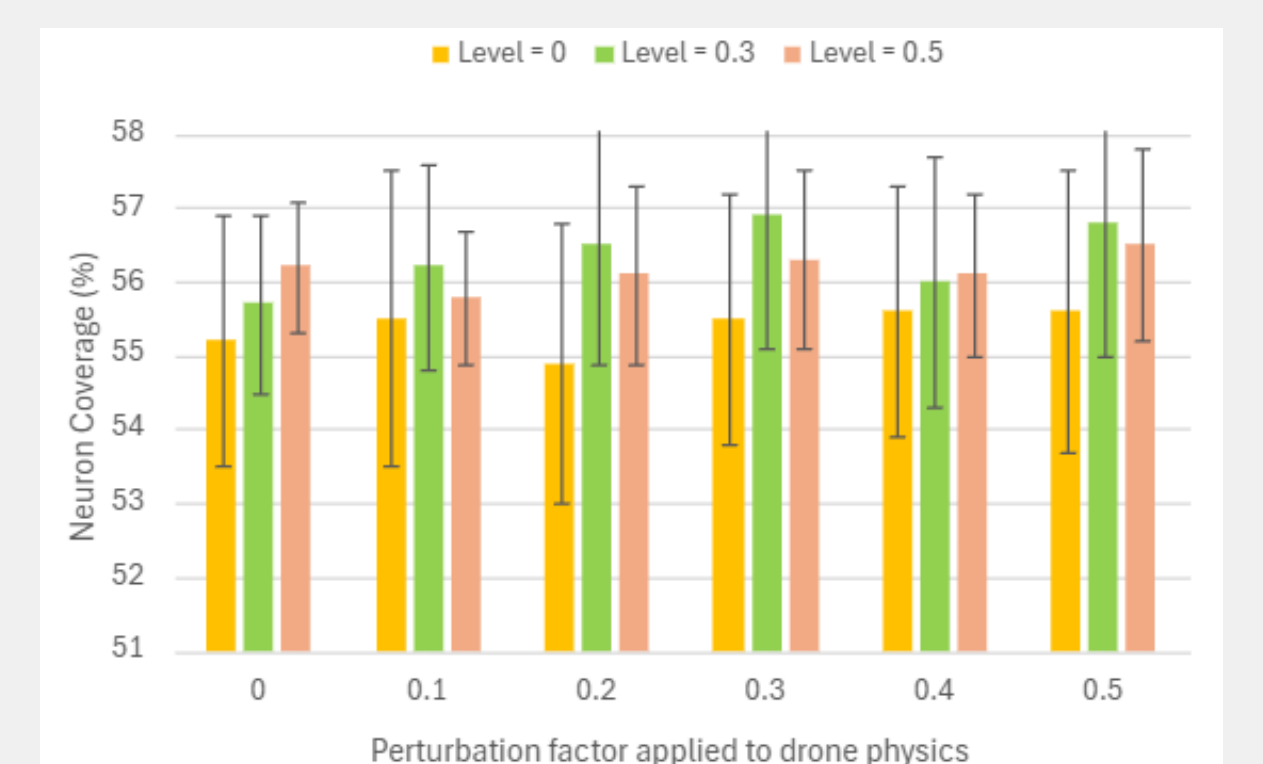


(b) INDI controller

Figure 6. Evolution of Reward



(a) End-to-end controller



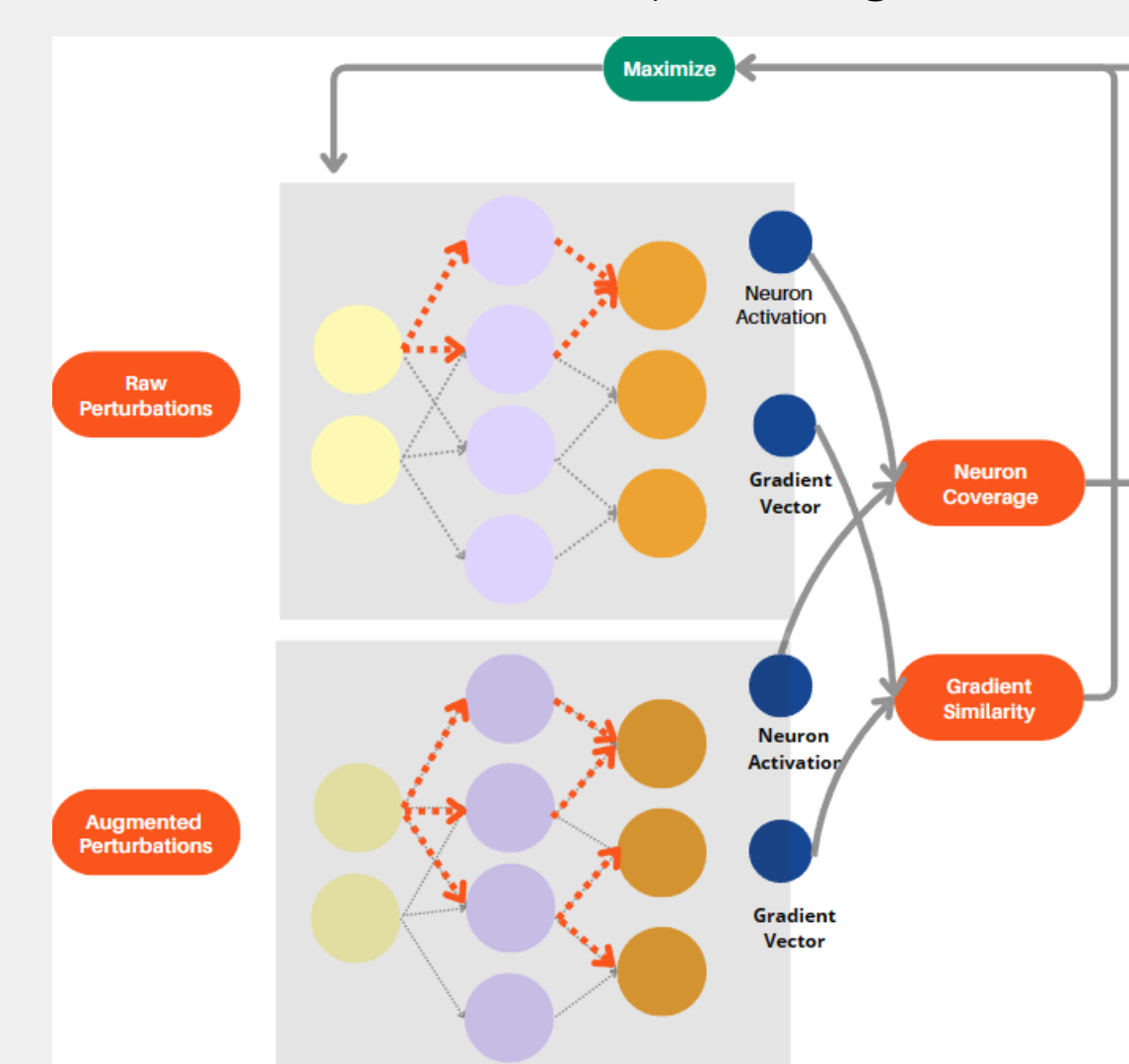
(b) INDI controller

Figure 7. Evolution of Neuron coverage

Despite promising results in the simplified scenario, our analysis hardly shows probant results here as it is very noisy. We can nonetheless notice that the drop in the reward when increasing the noise is mitigated for models trained with medium perturbations. Those also presents with a NC that is slightly higher than the others. This illustrates well the necessary trade-off in the amount of noise to implement in training. It is also interesting to notice that INDI RL has an overall better NC than End to End RL but also perform slightly worse rewards.

Future Work : A revisited Adversarial Learning framework

Instead of the traditional minimax formulation, we suggest a training setup with an adversarial agent choosing perturbations on the drone's physics (mass, wing span, etc) in order to maximize the agent's neuron coverage, rather than minimizing the same reward. This would probably ensure a higher Neuron coverage of the agent's network, ultimately leading to a more robust RL policy.



References

- Diedrich, Franklin W, and Joseph A. Drischler. Effect of spanwise variations in gust intensity on the lift due to atmospheric turbulence. 1957.
- Di Luca et al. Bioinspired morphing wings for extended flight envelope and roll control of small drones. *Interface Focus*, 2017.
- Enrico Ajanic et al. Bioinspired wing and tail morphing extends drone flight capabilities. *Science Robotics*, 2020.
- Enrico Ajanic et al. Sharp turning maneuvers with avian-inspired wing and tail morphing. *Communications Engineering*, 2022.
- Raid Rafi Omar Al-Nimaa et al. Robustness and performance of deep reinforcement learning. *Applied Soft Computing*, 105, 2021.
- Song et al. Autonomous drone racing with deep reinforcement learning. 2021.